

Code -

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%%writefile matmul.cu
#include <iostream>
#include <cuda_runtime.h>
using namespace std;

__global__ void matmul(int *A, int *B, int *C, int N) {
    int row = blockIdx.y * blockDim.y + threadIdx.y;
    int col = blockIdx.x * blockDim.x + threadIdx.x;

    if (row < N && col < N) {
        int sum = 0;
        for (int k = 0; k < N; ++k) {
            sum += A[row * N + k] * B[k * N + col];
        }
        C[row * N + col] = sum;
    }
}

int main() {
    int N = 512;
    int size = N * N * sizeof(int);

    int *h_A, *h_B, *h_C;
    cudaMallocHost(&h_A, size);
    cudaMallocHost(&h_B, size);
    cudaMallocHost(&h_C, size);

    int *d_A, *d_B, *d_C;
    cudaMalloc(&d_A, size);
    cudaMalloc(&d_B, size);
    cudaMalloc(&d_C, size);

    for (int i = 0; i < N; ++i) {
        for (int j = 0; j < N; ++j) {
            h_A[i * N + j] = i * N + j;
            h_B[i * N + j] = j * N + i;
        }
    }

    cudaMemcpy(d_A, h_A, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_B, h_B, size, cudaMemcpyHostToDevice);

    int threadsPerBlock = 16;
    int blocksPerGrid = (N + threadsPerBlock - 1) / threadsPerBlock;

    dim3 dimBlock(threadsPerBlock, threadsPerBlock);
    dim3 dimGrid(blocksPerGrid, blocksPerGrid);

    matmul<<<dimGrid, dimBlock>>>(d_A, d_B, d_C, N);
    cudaDeviceSynchronize();

    cudaMemcpy(h_C, d_C, size, cudaMemcpyDeviceToHost);

    // Print only 10x10 part
    for (int i = 0; i < 10; ++i) {
        for (int j = 0; j < 10; ++j) {
            cout << h_C[i * N + j] << " ";
        }
        cout << endl;
    }

    cudaFree(d_A);
```

```
    cudaFree(d_B);
    cudaFree(d_C);
    cudaFreeHost(h_A);
    cudaFreeHost(h_B);
    cudaFreeHost(h_C);

    return 0;
}
```

Output -

```
44608256 111586048 178563840 245541632 312519424 379497216 446475008 513452800
580430592 647408384
111586048 312781568 513977088 715172608 916368128 1117563648 1318759168
1519954688 1721150208 1922345728
178563840 513977088 849390336 1184803584 1520216832 1855630080 -2103923968
-1768510720 -1433097472 -1097684224
245541632 715172608 1184803584 1654434560 2124065536 -1701270784 -1231639808
-762008832 -292377856 177253120
312519424 916368128 1520216832 2124065536 -1567053056 -963204352 -359355648
244493056 848341760 1452190464
379497216 1117563648 1855630080 -1701270784 -963204352 -225137920 512928512
1250994944 1989061376 -1567839488
446475008 1318759168 -2103923968 -1231639808 -359355648 512928512 1385212672
-2037470464 -1165186304 -292902144
513452800 1519954688 -1768510720 -762008832 244493056 1250994944 -2037470464
-1030968576 -24466688 982035200
580430592 1721150208 -1433097472 -292377856 848341760 1989061376 -1165186304
-24466688 1116252928 -2037994752
647408384 1922345728 -1097684224 177253120 1452190464 -1567839488 -292902144
982035200 -2037994752 -763057408
```